DEPI · DATA ANALYSIS TRACK · FINAL PRESENTATION

RailSmart

A UK Railway Intelligence Platform

31,653 journeys | Jan – Apr 2024 | 65 routes | Python · SQL · Tableau · Power BI · Web · Mobile · AI

Khaled Eissa · Elsayed Elgohary · Rana Yasser · Rahma Nasser · Ahmed Shaaban

THE PROBLEM

Passengers are owed money.
Almost none of them ask for it.

UK rail passengers whose train is delayed or cancelled are entitled to compensation.
But claiming it is manual, fiddly, and easy to forget — so most people never do.

In this dataset alone, across just four months:

13.2%
of journeys disrupted
4,172 delayed or cancelled

73.2%
never claimed a refund
3,054 entitled passengers

UNCLAIMED COMPENSATION

£133,568

18% of total revenue · 3.45× what was actually refunded

That gap is the reason
RailSmart exists.

THE SOLUTION

One cleaned dataset, driving every surface

No figure is retyped. The same verified table feeds four visual surfaces and three passenger-facing roles.

RAW
railway.csv
19 cols · 31,653 rows

→

CLEAN + ENGINEER
railway_cleaned.csv
42 cols · notebook + SQL

→

- Python analysis
- Tableau
- Power BI
- Web app
- Mobile app
- AI models

Role: Data Engineer · Role: Passenger · Role: Station Manager

The dataset

31,653
journeys

65
routes
12 origins · 32 destinations

121
days of history
1 Jan – 30 Apr 2024

19→42
columns after engineering

Ticket, route, scheduled vs actual arrival, journey status, price, refund flag. State the date range aloud — it bounds every claim in this deck.

IMAGE

Data dictionary table, or a first-look `df.head()` screenshot from the notebook

landscape · ~4:3

Data quality — what was actually broken

The credibility slide. Four real defects found and fixed — the last one is ours.

1 · The `'None'` → NaN trap

20,918 blank railcards filled as **'No Railcard'**, deliberately not `'None'` — which pandas re-reads as NaN and would corrupt the file on reload.

2 · Colliding delay labels

Signal Failure / Signal failure, Weather / Weather Conditions logged separately — splitting every root-cause count. Consolidated before counting.

3 · Nulls left as null

1,880 cancelled journeys have no arrival time. That NaN is the **truthful value** — the train never arrived — not a gap to fill.

4 · The fix that broke the next chart

`.notna()` then flagged **all 31,653** as railcard holders → the chart printed £nan. We caught our own regression. Real adoption: 33.9%.

Feature engineering — 19 → 42 columns

Every derived column earns its place by answering an analysis question. The six that matter most:

- **Delay_Minutes** — scheduled vs actual, the spine of every punctuality metric
- **Days_In_Advance** — booking lead time, powers the urgency-tax analysis
- **Time_Of_Day** — morning peak / midday / evening bands
- **Route** — origin → destination pair, the unit of reliability
- **Is_Weekend · Price_Per_Minute** — demand and value normalisers

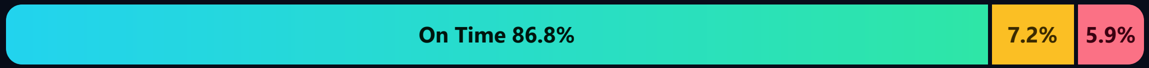
IMAGE

Feature-engineering code cell, or a correlation heatmap of the new columns

landscape · ~4:3

Punctuality baseline

SERVICE MIX · 31,653 JOURNEYS



• On time 27,481 • Delayed 2,292 • Cancelled 1,880

The number that matters is the combined **13.2% disruption rate** — because that is the population entitled to compensation.

IMAGE

Journey-status pie or bar from the notebook (section 9.1)

landscape · ~4:3

Where the money leaks

£38,702
refunds actually paid

£133,568
left unclaimed
3.45× larger

Claim rate by status · **Cancelled 30.4%** **Delayed 23.8%**
By channel · **Online 28.6%** **Station 25.8%**
No on-time journey ever claimed — the entitlement lives entirely inside the 13.2%.

IMAGE
The unclaimed-compensation bar pair from notebook section 10.4
landscape · ~4:3

Why trains are late

Weather leads — **not** signal failure. Share of 4,172 disrupted journeys.

Counts: 1,372 · 970 · 809 · 707 · 314

IMAGE

Delay-cause chart from notebook section 9.2 (or the Tableau delay sheet)

landscape · ~4:3

Route reliability

Worst routes by on-time rate — minimum 30 journeys, so noise doesn't top the table.

ROUTE	JOURNEYS	ON-TIME
Edinburgh Waverley → London Kings Cross	51	0.0%
Liverpool Lime Street → London Euston	1,097	19.9%
Manchester Piccadilly → London Euston	345	29.6%
Birmingham New St → Manchester Piccadilly	224	49.1%

Some routes record only cancellations, never delays — shown as such, not as "0 min delay".

IMAGE

Route-reliability bar from notebook 10.2, or the Station Manager delay-watch panel

landscape · ~4:3

The urgency tax

£17.48

planned (>2 days)

3,450 bookings

£24.17

urgent (≤2 days)

28,203 bookings

+38.3% paid for leaving it late

Honest caveat, stated first: median lead time is **1 day** and nothing is booked beyond 28 days — so the "urgent" bucket holds most of the data.

IMAGE

Urgency-tax bar chart from notebook section 10.3

landscape · ~4:3

From findings to actions

Every recommendation carries a number.

Automate refund claims

Detect entitlement and file automatically. Directly addresses the 73.2% who never claim.

Recover £133,568 / 4 months

Weather-linked staffing

Weather is the #1 disruption cause. Pre-position crews on the 5 worst routes ahead of forecasts.

Targets 32.9% of disruptions

Nudge station-channel buyers

Station buyers claim 3 pts less often than online. A prompt closes the gap.

25.8% → 28.6% claim rate

Forecasting — and an honest result

DAILY MAPE	Flat mean	Seasonal naïve	SARIMA
Rides	18.42%	22.24%	**18.21%**
Revenue	**20.45%**	23.21%	21.65%

SARIMA barely beats a flat mean on rides and loses on revenue. Demand is stationary — no trend, weekday swing only ±11%. The honest forecast is the mean with a 95% interval. Reporting that beats inventing accuracy.

IMAGE

30-day backtest / forecast chart with prediction interval, notebook section 14

landscape · ~4:3

Predictive models

FARE ESTIMATE

£2.75 MAE

R^2 0.963. 79% of the signal is the **route**, only 5% is booking lead time.

DELAY RISK

0.979 ROC AUC

Recall 0.657. Delays are **structural** — same routes, same hours, repeatedly.

CLAIM PROPENSITY

0.860 ROC AUC

Recall 0.737. Trained on the 4,172 **entitled** only — who actually claims.

IMAGE

Feature-importance charts, or the end-to-end model run from notebook section 13

wide · ~16:6

Honest model evaluation

Accuracy is a trap on rare events — we present the baseline ourselves.

MODEL	Baseline	Accuracy	Recall	AUC
Delay risk	92.8%	96.3%	0.657	**0.979**
Refund (all)	96.5%	97.3%	0.304	0.744
Claim (entitled)	73.2%	75.8%	0.737	**0.860**
Cancellation	94.1%	—	0.000	0.726

Pooling delay + cancellation drops AUC 0.98 → 0.84. So we ship delay as a prediction, cancellation as a route statistic.

IMAGE

Confusion matrices from notebook section 13 (delay, claim propensity)

landscape · ~4:3

AI vision — platform crowd counting

A pretrained **YOLOv8** model counts people on the platform in real time — boxes and a live count drawn on each frame.

- Ultralytics YOLOv8, runs on real CCTV footage
- Feeds the passenger "how busy is my station" view
- An extension **beyond** the tabular rubric — labelled as such

IMAGE / VIDEO FRAME

A frame from docs/person_count_frame.png with detection boxes + live count

landscape · 16:9

Four surfaces, one dataset

The same cleaned table drives all of them, with no reshaping.

TABLEAU

KPI Overview dashboard screenshot

POWER BI

Report page screenshot

WEB APP

Data Engineer dashboard screenshot

MOBILE

Expo app screenshot

Live demo — the web platform

- **Data Engineer** — KPIs, regional revenue, 3D train
- **Passenger** — journey planner, fares, station crowding
- **Station Manager** — fleet, maintenance, delay watch

QR

Drop the live Vercel QR here.
Keep the recorded demo cued as a fallback.

SCREENSHOT

Web app — the Data Engineer dashboard with the 3D train

landscape · 16:9

The mobile app

All three roles, mirrored on the phone in **Expo** — plus a full book → pay → manage flow.

- Same real dataset as the web app
- Charts drawn in native views — Expo Go compatible
- **Live via** `expo start --tunnel` — works off-network

SCREEN 1

Passenger tab

SCREEN 2

Operations tab

Limitations — stated up front

Honesty is scored, and it pre-empts half the panel's questions.

- **121 days** of history — captures a weekly cycle, not a yearly one
- **Single dataset**, one source — no external validation
- **No live feed** — surfaces read pre-exported JSON, not a real-time API
- Models are **first-pass baselines**, not tuned for production
- Cancellations are **not predictable** from schedule data alone
- Booking advice can't extend past **28 days** — the data's own limit

Bringing RailSmart to Egypt

For the benefit of public transport at home — honestly scoped.

What transfers directly

The cleaning pipeline, feature engineering, the delay and claim models, and the three role dashboards. None of it is UK-specific.

What does not

UK Delay Repay is a statutory scheme. Egypt has no equivalent — so the refund feature becomes a **policy proposal**, not a shipped one.

What we'd need

12 months of ticketing with scheduled **and** actual times, at journey grain. Start with one high-density corridor, not the whole network.

The analysis is portable; data access is the hard part. No ridership figures quoted without a source.

CONCLUSION

A data product with a number behind every claim

£133,568 unclaimed — the case for RailSmart	**86.8%** on-time, from 31,653 journeys	**0.979** delay-model ROC AUC	**6** delivery surfaces built

Thank you — questions welcome.

REPO QR

Eissa · Elgohary · Yasser · Nasser · Shaaban · DEPI Data Analysis